Algebra 1
Unit 9 – Part 2
Assessment

Name____________________
Date_________________
Period__________

Solve each equation. Show your work in the space provided.

1. $x^2 - 4x - 12 = 0$

Work	
Solutions	$x =$ ____________ $x =$ ____________

2. $-5(x+4)^2 + 19 = -6$

Work	
Solutions	$x =$ ____________ $x =$ ____________

3. $2x^2 + 12x + 15 = 0$

Work	
Solutions	$x =$ ______________ $x =$ ______________

4. $x^2 - 2x - 28 = 7$

Work	
Solutions	$x =$ ______________ $x =$ ______________

5. A data scientist is comparing real estate market trends in the southern United States for the years 2005 to 2020. The equation, $y = 6(x - 8)^2 + 192$, represents the number of housing units sold, y, in thousands, for each year since 2005, x.

a. Write the constraints for the real estate market in the southern United States. Show your work or explain how you determined the constraint.

Variable	**Constraint**	**Work or Explanation**
x		
y		

b. Determine when the number of housing units sold in the southern United States is $216{,}000$. Show your work.

Work	
Year(s)	

6. The US government monitors the consumption of different products. The table shows y, the amount of ice cream consumed, in millions of pounds, for x years since 2010.

x	0	1	2	3	4	5	6	7	8
y	4,354	4,222	4,114	4,030	3,970	3,934	3,922	3,934	3,970

The quadratic equation that models the amount of ice cream consumed, in millions of pounds, since 2010 is shown.

$$y = 12(x - 6)^2 + 3922$$

Determine when the amount of ice cream consumed in the United States would be 5,650 millions of pounds.

Work	
Year(s)	